



IMU-augmented Patient-related Outcome Measure for Knee Arthroplasty Patients

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Received: 26 May 2025 / Accepted: 14 August 2025 / Published online: 31 August 2025
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Abstract

Purpose The aims of this study are (1) to derive a set of time series features to predict the total knee contact force during gait from two ankle-mounted inertial measurement units (IMUs); and (2) to explore the best combination of PROMs and IMU-derived, biomechanical measures for the prediction of postoperative quality of life (QoL) and reduce the inconsistency between biomechanical and PROMs.

Methods Synced motion capture (optical and inertial) data were collected from four healthy participants to obtain a suitable time-series feature set that relates to knee contact forces (KCF). Then, using data from 28 patients during overground walking, we generated linear and random-forest regressor models for estimating QoL. These models were evaluated using temporal cross-validation.

Results Overall, features from the IMUs could predict the total KCF ($R^2 > 0.90$, RMSE = 0.14, %BW). A machine learning model was trained on the data obtained during the recovery (up to 1 year) to predict EQ-5D-5 L score (patient QoL). The models were able to predict within a margin of 1.81–4.40 units of the actual score 95% of the time.

Conclusion The presented study derives surrogate measures of total knee contact force (KCF) using ankle-mount IMUs. We identified the best combination of PROMs and IMU-derived measures for postoperative QoL at four time-points. The optimal metrics varied for each time point: OKS only, knee kinematics with OKS, KCF with OKS, and KCF only. Further data collection will be required to compile a comprehensive dataset to generate a robust predictive model.

Highlights

- 20% of knee replacement patients will experience dissatisfaction. Quality of life not explained by existing outcome assessments used in the clinic. Machine learning approach can relate wearable data to outcome assessments. Hybrid score removes subjectivity and patient misinterpretation of the question.

Keywords Machine learning · PROMs · Knee replacement · IMU

1 Introduction

Surgical knee replacement is the most effective option for patients with end-stage knee osteoarthritis (OA) to relieve pain and restore function. The demand for arthroplasty surgery is predicted to rise significantly, and post-operative outcome assessment is essential to monitor patient progress. This is important as 20% of patients will experience dissatisfaction, leading to lower quality of life [1, 2], which can lead to surgical revision if the implant fails. Clinicians typically use patient-related outcome measures (PROMs). These are standardised, self-reported questionnaires intended to reflect the patient's perception of function and well-being.

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PROMs commonly used for the knee are: (1) Oxford Knee Score (OKS) and (2) Forgotten Joint Score (FJS).

OKS is a PROM system widely used in research and clinical applications [3, 4], and is increasingly used in New Zealand, England, and Sweden [5]. It is a 12-item questionnaire assessing knee symptoms and function, and it is intended to be short, practical, reliable, valid, and sensitive to clinically significant change relative to scales such as the SF-36 and Arthritis Impact Measurement Scale (AIMS). The score has been designed to be entirely patient-derived and has been validated across different languages [6–10]. However, some issues with clarity and responsiveness have been noted, including a considerable ceiling effect [4, 11], making it less suitable for investigating small differences in patients with good or excellent postoperative results. Furthermore, the low OKS at six months, five, and ten years post-surgery have been found to be associated with knee replacement revision surgery within two years of the OKS taken [12].

FJS is a 12-item questionnaire based on the patient's awareness of the artificial joint in daily living. It includes parameters measuring whether the joint is free of pain, provides a sufficient range of movement, and has stability in all degrees of flexion during activities of daily living (ADLs). Early studies have found the FJS to have good construct validity and test-retest reliability, correlation with the OKS, and lower ceiling effect and responsiveness than the OKS [3, 5, 13].

However, these knee-specific scores are still often dominated by pain, which masks functional changes. This could explain the poor correlation between PROMs and objectively assessed physical function, leading to poor estimation of patients' Quality of Life (QoL) and patient satisfaction.

The EQ-5D-5 L is the 5-levelled version of the EQ-5D introduced by the EuroQol Group in 2009 for measuring health-related quality of life (HRQoL). The system comprises five dimensions (mobility, self-care, usual activities, pain/discomfort, and anxiety/depression), each with five levels (no problem, slight problems, moderate problems, severe problems, and extreme problems), defining a total of 3,125 unique health states [14, 15]. It is more suitable for knee OA as it is better at recognising small-to-moderate health status differences [16, 17]. It is also a widely accepted measure of pain that is available with the visual analogue scale (EQ-5D-5 L VAS), where the patient marks their health state on a graphic sliding scale between 0 (worst imaginable health state) and 100 (best imaginable health state). This allows it to measure a broader range of health and is closer to the patient's perspective rather than population-based value sets [18, 19]. The EQ-5D-5 L was also able to measure patient dissatisfaction not explained by knee-specific PROMs because it is less influenced by subjectivity, socioeconomic, or physiological factors.

A biomechanical metric that represents knee joint loading could be used to provide a measure of the patient's knee function. These forces are important in the first six weeks of recovery for muscle and bone growth to support the long-term success of the implant and avoid complications such as aseptic loosening [20–22]. Wearable sensors such as inertial measurement units (IMUs) could be used to estimate this loading and are becoming more popular as they are cheap, accessible, and yield results that are highly comparable to those obtained through OMC [23–25]. Furthermore, they can capture out-of-clinic motion data without clinician supervision.

This study aims to explore the best combination of PROMs and IMU-derived biomechanical measures for the prediction of postoperative QoL and indirectly estimate patient satisfaction during the recovery period.

2 Materials and methods

2.1 Data Collection for Creating and Evaluating the Surrogate Patient QoL Models

Knee replacement surgery patients were recruited with written informed consent for this study under ethics approval from the University of Auckland Human Participants Ethics Committee. 28 patients (13 females and 15 males; 23 TKR and 5 UKR, age: 68.2 ± 6.2 ; height: 1.67 ± 0.11 m; body mass: 87.5 ± 19.4 kg; BMI: 31.3 ± 5.4 kg/m²) were considered for this study. Patients attended physiotherapy sessions twice a week during the six-week recovery phase, with at least one session involving IMU tracking (Fig. 1) using the IMeasureU BlueTrident IMUs attached at the ankles. Acceleration and angular velocity data were collected along with the OKS, EQ-5D-5 L VAS, and FJS online. The relative OKS was calculated as the difference between the current and preoperative OKS. Follow-up questionnaires were administered at three, six, and 12 months post-surgery via phone or email. Due to communication challenges after physiotherapy sessions, response rates varied. The available EQ-5D-5 L scores and details of each dataset per timepoint are shown in Table 1. Gait lab session pre- and post-surgery (at 6 weeks) was optional for the patients, and the result was presented as a separate study [25].

At physiotherapy sessions, patients followed the exercise plan as shown:

- Speed walking for 50 m.
- Cycling for 3 min.
- Lunges $\times 10$: patients placed their operated leg onto the second step and leaned forwards, bending their operated

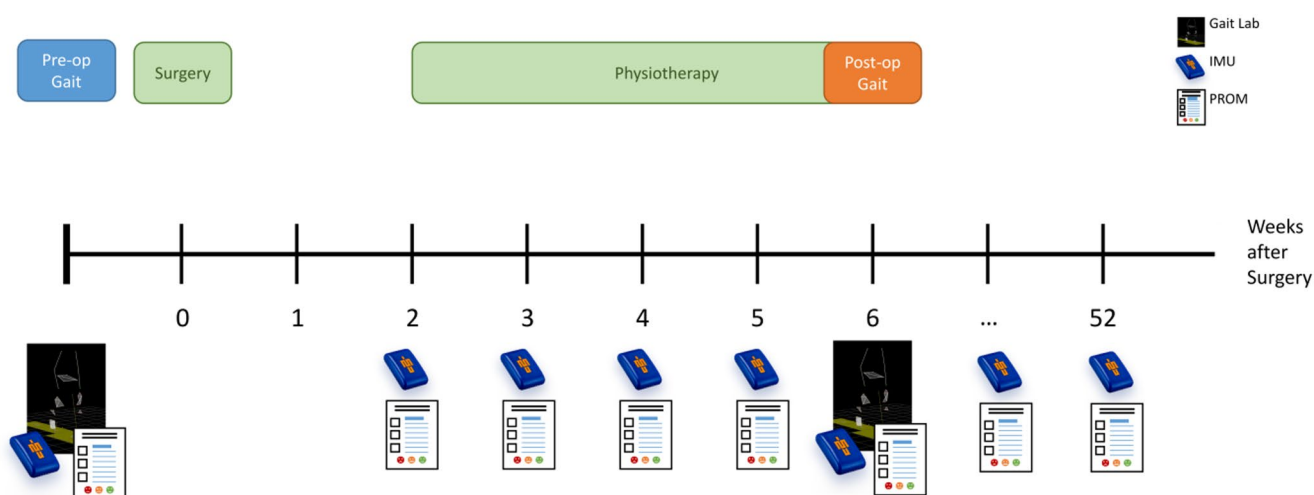


Fig. 1 Study timeline following preoperative screening, postoperative physiotherapy and treatment. Patients were contacted to answer post-operative PROMs at the checkpoints of 3 months, 6 months, and 1 year after surgery

Table 1 Summary of the patient dataset for week 6, month 3, month 6, and year 1. See appendix A.2 for patient QoL scores across all post-operative timepoints

	Physio 2	Week 6	Month 6	Year 1
No. datapoints available	28	18	16	11
Age	68.2 ± 6.2	67.8 ± 6.3	69 ± 6.1	68.5 ± 5.9
Height	166.7 ± 10.7	167.1 ± 11.3	165.8 ± 11.3	167.5 ± 8.9
Weight	87.5 ± 19.4	90.4 ± 19.1	88 ± 19.3	81.2 ± 20.6
BMI	31.3 ± 5.4	32.2 ± 4.8	31.8 ± 4.8	28.8 ± 6.5

knee, reaching for the wall, and holding the position for 10 s.

- Steps up and downs $\times 20$: patients stepped up onto the first step, leading with their operated leg, and stepped back down, leading with their non-operated leg.
- Knee extensions $\times 20$: patients straightened their operated leg while keeping their thigh on the seat, held for 3 s.
- Sit to stand $\times 20$: patients stood up from the chair without using their hands 20 times.
- Wobble board for 2 min: patients stepped onto the wobble board, with their feet hip distance apart and remained in the position (they could hold the bar for support when needed).
- Calf rises $\times 20$: patients faced the bar, rose onto tiptoes and held for 3 s before lowering down.
- Stretches $\times 3$: a staff member helped patients to stretch the calf, hamstrings, and quadriceps for 20 s.
- Kneeling $\times 5$: patients stepped their non-operated leg forward to help support themselves, knelt onto their operated knee and held for 10 s before standing up (only from weeks 3 to 6).
- Stair climb (only from week 4 to week 6).

- Leg press: patients placed their operated leg onto the front plate, pushed back slowly (not fully straightened) and held for 3 s before returning to the starting position with a weight of 30 pounds (week 2) to 40 pounds (week 3 to 6).
- Hamstring curl $\times 20$: patients were seated in the chair with the strap around the ankle of their operated leg, and they slowly pulled backwards, held for 3 s, and slowly released forwards with a weight of 2.5 kg (week 1) to 5 kg (week 2 to 6).

2.2 Patient QoL Surrogate Modelling

The IMU signals were used as an alternative for OMC to provide an objective, biomechanical contribution to the prediction. By using feature engineering, various sets of IMU features corresponding to different biomechanical variables were assessed against patient satisfaction (Fig. 2). These feature sets were:

1. KCF feature set [26]: this corresponds to the chosen, generalised KCF prediction model that uses left and

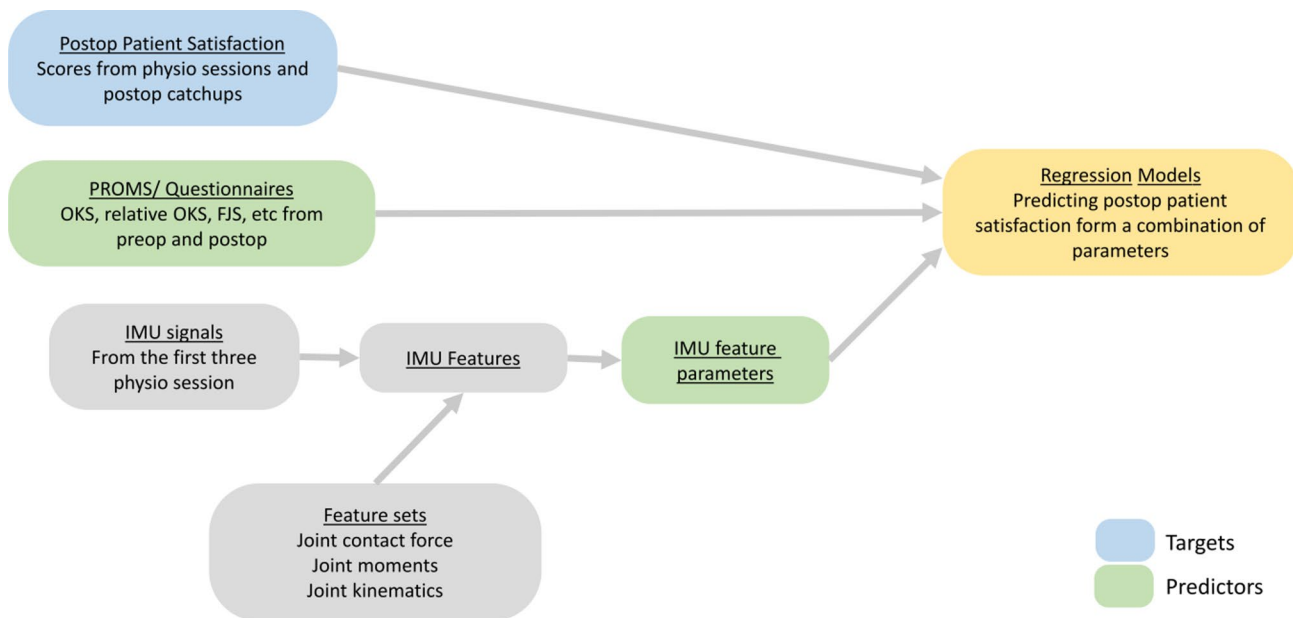


Fig. 2 Overview of the variables contributing to the patient QoL prediction model

right ankle IMUs, and triaxial accelerometer data. Feature extraction of the top 50 features using a window size of 0.6 s.

2. KAM feature set [26]: obtained using the results of inverse dynamics via the pipeline from uses left and right ankle IMUs, and triaxial accelerometer data. Feature extraction of the top 50 features using a window size of 0.6 s.
3. Kinematics feature set: obtained by Cantamessa et al. [27] for knee flexion predictions using a workflow similar to the workflow presented in Fig. 1. Uses left and right ankle and thigh IMUs, triaxial accelerometer and gyroscope data. Feature extraction of the top 100 features using a window size of 0.6 s [28, 29].

To evaluate the models, temporal cross-validation was carried out. The first 80% of each participant's data was used for training, and the further 20% was reserved for model evaluation. The coefficient of determination or model fit (R^2), the root mean square error (RMSE), and the mean absolute error (MAE) were calculated and compared as validation metrics.

3 Results

3.1 Patient QoL Scores

The EQ-5D-5 L score was associated with various combinations of different parameters using linear regression (LR) and Random Forest Regression (RFR). Figure 3 shows the

LOOCV outcomes when associating EQ-5D-5 L with various PROMs obtained during physiotherapy. The greatest variability was found across timepoints, with the predictions for physio 2 showing the lowest MAE and SD when outliers were included. When the outliers were excluded, there was slightly less variability across timepoints, but it is still noticeable. As well as this, year 1 shows a reduced MAE and SD, lower than that of physio 2. The predictions when using different PROMs as predictors (including the combination of PROMs and IMU-derived biomechanical measures) yielded similar outcomes. Generally, the OKS and FJS showed slightly better MAE values but displayed significant overlap in SDs. The most noticeable difference in the models was seen when using all three measures (combined), which shows greater MAE at week 6, month 6, and year 1.

3.2 Estimating Patient QoL with IMUs

The LOOCV results for the four chosen models are shown in Fig. 4. Month 6 and year 1 showed the least errors in their predictions. Furthermore, they had the most consistency in accuracy across all participants. Conversely, physio 2 and week 6 showed the greatest errors in their predictions. Some predictions showed greater inaccuracies, such as P012, P030, and P043 for physio 2, and P026, P039, and P042 for week 6. Two of the three inaccurate data points for week 6 were the lowest patient QoL scores seen in the data set and were overestimated.

Figure 5 illustrates the mean and 95% confidence intervals of the difference between the actual and predicted

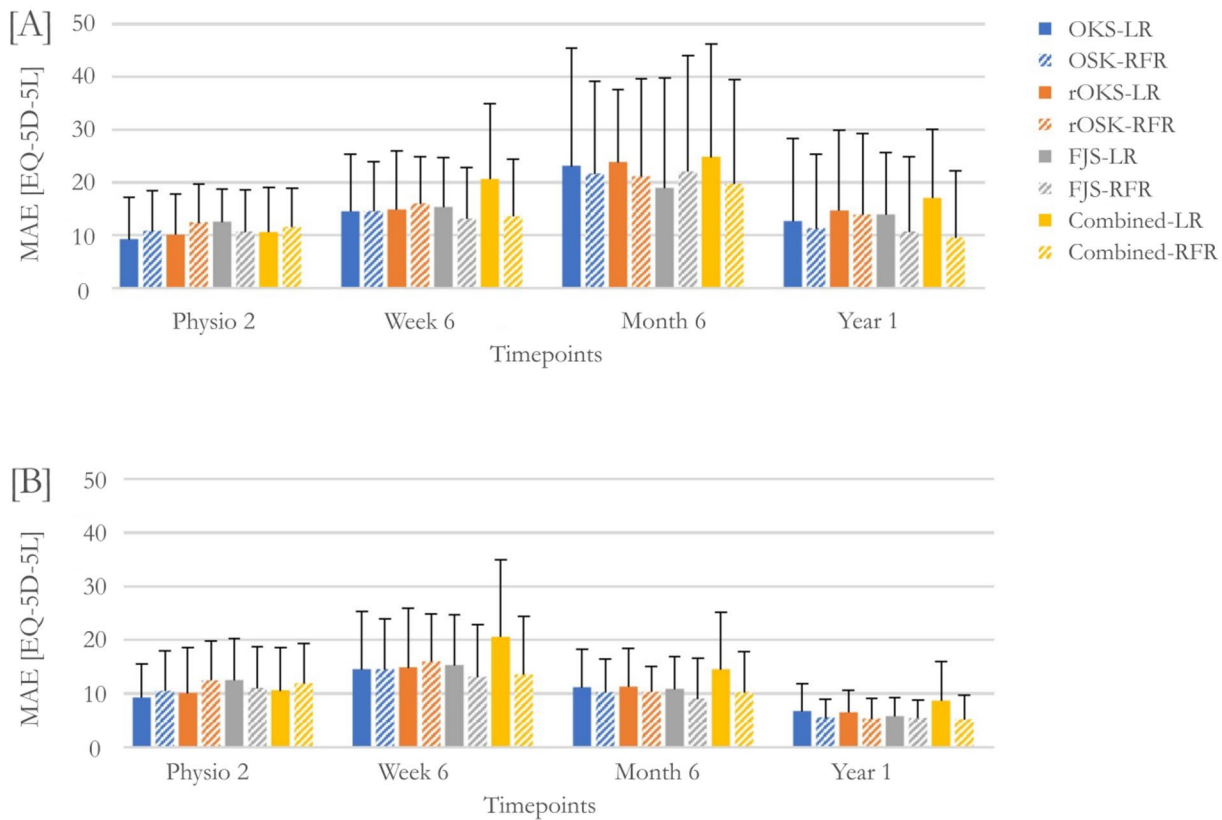


Fig. 3 MAE values from the LOOCV analysis of patient QoL prediction models using different PROMs as predictors. [A]: Including outliers; [B]: excluding outliers

patient QoL scores calculated from a zero-dimensional SPM paired t-test. The paired t-test yielded p-values of 0.513, 0.182, 0.239, and 0.621 for physio 2, week 6, month 6, and year 1, respectively. As they were all greater than 0.05, the null hypothesis of no difference between the predicted and actual patient QoL scores could not be rejected.

4 Discussion

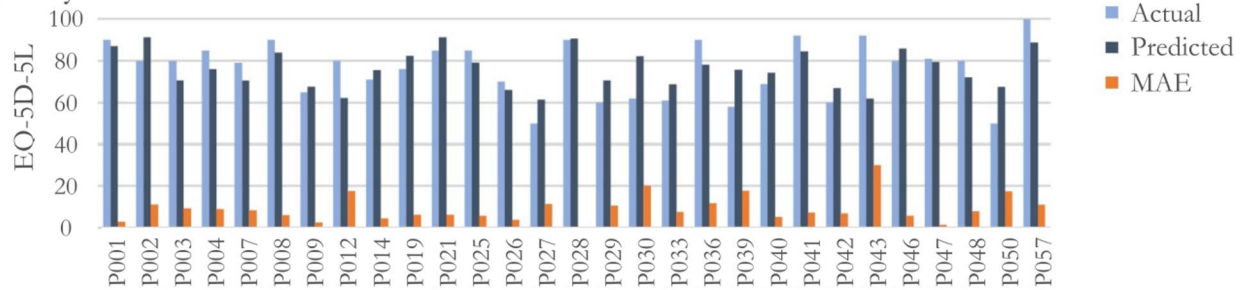
This study aimed to investigate the use of time series features from IMU sensors calibrated to gait biomechanics (total KCF and joint kinematics) together with existing PROMs for estimating patient QoL (EQ-5D-5 L VAS). The selected IMU-derived biomechanical features were then combined with PROMs to predict postoperative patient QoL. The best combination of metrics varied at each timepoint. The models were able to be predicted within a small point margin 95% of the time for each timepoint, with the best model being trained on IMU features based on KCF.

4.1 Surrogate Modelling

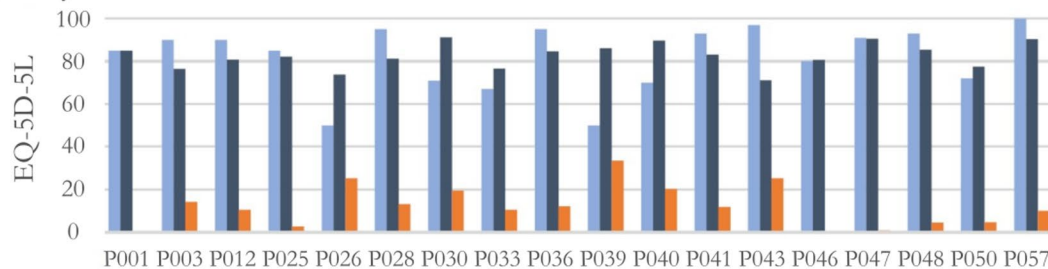
The study also explored the use of different combinations of PROMs and IMU-derived biomechanical measures to determine the best combination to predict postoperative patient QoL. Various PROMs and IMU features were associated with the EQ-5D-5 L VAS as a representation of patient QoL during physiotherapy, and at six weeks, six months, and one year post-surgery. It was found that the best combination of metrics varied at each timepoint, with them being OKS only, knee kinematics with OKS, KCF with OKS, and KCF only for each respective timepoint. There weren't any statistically significant differences between the predicted and actual patient QoL scores. The chosen models were able to be predicted within a margin of 4.40, 3.62, 2.14, and 1.81 units of the actual score at 95% of the time for each timepoint, respectively.

The association of KCF with the IMU done in this study is an extension of what has been done previously [28, 30]. Their methods used the net knee reaction force computed through inverse dynamics as a surrogate for the total KCF. In this study, we use total KCF that arises from the contraction of muscles around the knee joint and is a more accurate

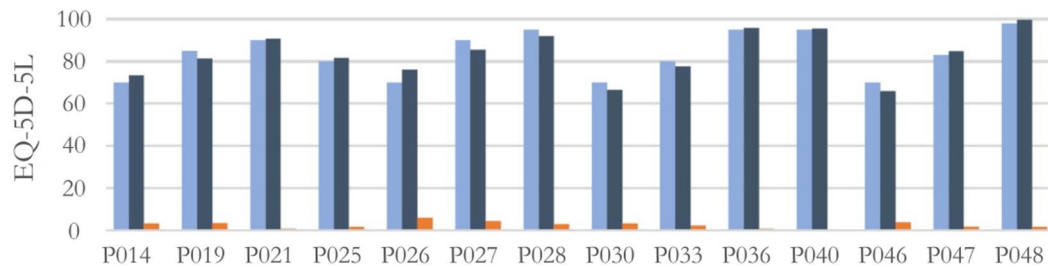
[A]: Physio 2



[B]: Physio 2



[C]: Month 6



[D]: Year 1

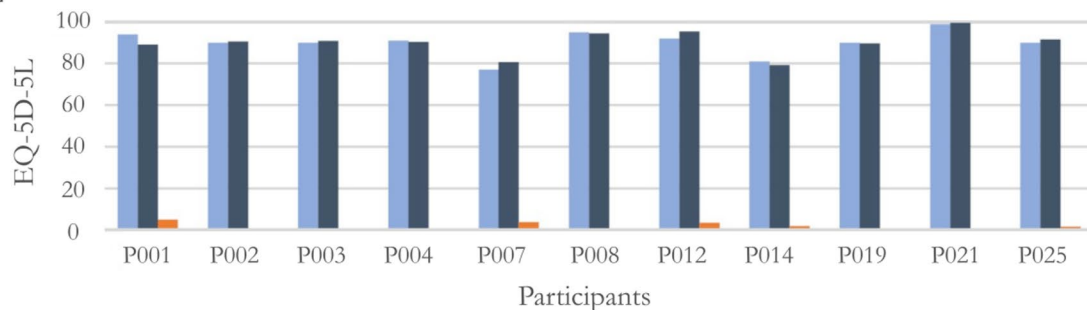


Fig. 4 Comparison of the predicted and actual patient QoL scores of each chosen model. [A]: Physio 2, [B]: Week 6, [C]: Month 6, [D]: Year 1

representation of the loading at the knee. Furthermore, they did not use feature engineering to predict the KCF across the stance phase. Instead, they predicted only the KCF at heel strike using the peak tibial acceleration via LR. When looking at the top features used for training, the maximum

acceleration was not as highly relevant, and other features dominated the list.

When examining the IMU-derived biomechanical feature sets, knee kinematics were initially the best metric associated with patient QoL during physiotherapy, but the KCF became more significant from week six onwards. This

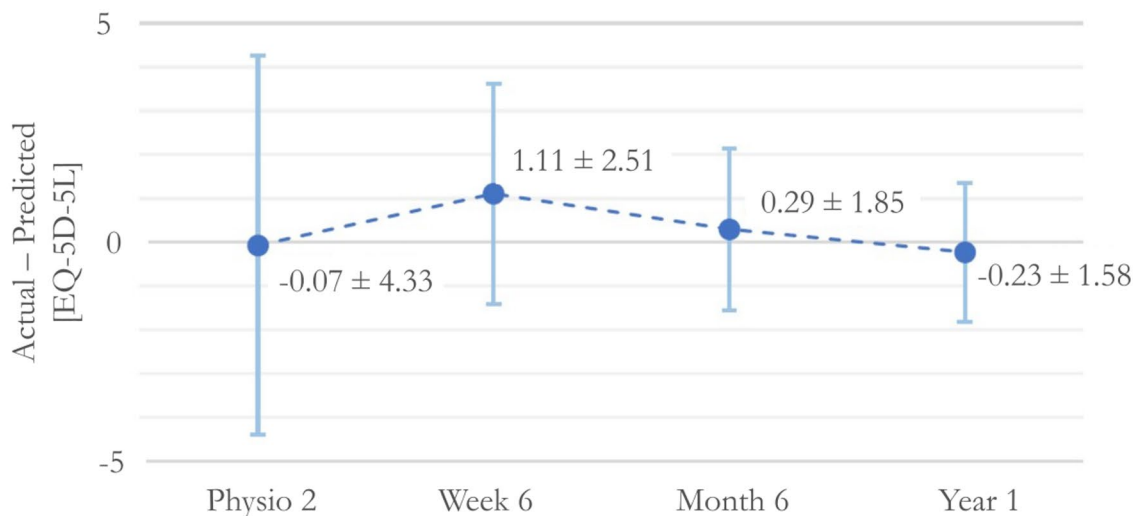


Fig. 5 The mean difference and 95% confidence interval between the actual and predicted patient QoL across the four timepoints

suggests that a multimodal approach is likely the best predictor as patients change their behaviours across rehabilitation. Importantly, both the patient kinematics and KCF, which improved the prediction of patient QoL across 6 weeks at different times, are functional measures from the IMU that add value over the standard OKS. This added value could, for example, provide the necessary information to compare different implant devices, where previous studies in the literature often show no difference [31–33].

4.2 Limitations, Assumptions, and Benefits

The models are built for specific time ranges (recovery periods) and are not one model fits all. This approach will have better accuracy and reliability due to reduced variation in the recorded IMU data. This makes sense as patients immediately perceive improved mobility as part of recovery. In comparison, the KCF is not directly perceivable but is important in promoting muscle and bone growth that supports the long-term success of the implant [21]. Additionally, the KAM feature set was not found to perform the best at any time. This makes sense as we expect the KCF to outperform the KAM as it is a direct measure of loading. Previously, the KAM was shown not to be a good predictor of KCF [34].

Three outliers were identified in the training dataset at month 6 and year 1 and were excluded from the analysis for various reasons. For example, one patient's (P009) lower patient QoL score was likely to be due to pain experienced from the patient's recent hip surgery and so can be considered a case outside of the scope of this investigation. As for

the other two patients (P029 and P039), no reasons have been discovered to explain the lower patient QoL scores at month 6. As their scores after one year have yet to be collected, we were also unable to investigate whether there is an ongoing effect. Because of this, these data points could not be ignored as they could be representative of the spread of patient QoL scores.

The LR algorithm showed better accuracy compared to the RFR across all timepoints. This aligns with the findings by Huber et al., [18], who found extreme gradient boosting, linear model, multistep elastic net, and neural net methods to show the best performance in predicting PROMs by a small margin.

The predicted score from these algorithms also has the benefit of not having the patient answer a questionnaire, removing subjectivity and the possibility of the patient misinterpreting the questions from the assessment. And due to the use of the same feature set as the kinematics and kinetics models, we could estimate the patient's gait parameters if their score suggested more information is needed to assess their joint function.

5 Conclusion

The outlined workflow was able to derive surrogate measures of the total KCF from two ankle-mount IMUs. From regression modelling, we were able to determine the best combination of PROMs and IMU-derived biomechanical measures with postoperative patient QoL at four timepoints (during physiotherapy, six weeks, six months, and one

year after surgery). It was found that the best combination of metrics varied at each timepoint, with them being OKS only, knee kinematics with OKS, KCF with OKS, and KCF only for each respective timepoint. With the continuation of data collection in the study, a sufficient dataset including low QoL scores should be compiled for further analysis.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s40846-025-00974-z>.

Acknowledgements We would like to acknowledge Science for Technological Innovation for funding this project, and we would also like to acknowledge VICON-IMeasureU for their technical support for this project.

Author Contributions All authors contributed to the study conception and design. Sabina Yang, Shasha Yeung, Faseeh Zaidi, Sebastian Weaver, and Megan Lovatt performed material preparation and data collection. Ted Yeung and Sabina Yang performed data analysis. Ted Yeung and Sabina Yang wrote the first draft of the manuscript, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding Open Access funding enabled and organized by CAUL and its Member Institutions. Funding from the Science for Technological Innovation (NZ), National Science Challenge Seed Project (UOAX2005).

Data Availability Data can be made available on request.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics approval Approved by the Auckland Health Research Ethics Committee on 04/03/2019 for three (03) years. Reference number 000072.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent to publish The authors affirm that human research participants provided informed consent for publication.

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